

Math515 - Financial Mathematics II

Spring 2014

Final Exam

You must show all of your work to get a credit for a correct answer.

1. A fair coin is thrown n times. Show that the conditional probability of a head on any specified trial, given a total of k heads over the n trials, is k/n ($k > 0$).
2. Let X and Y are any two random variables. Show, by an example, that we can have $\text{Cov}(X, Y) = 0$ but X and Y are not independent.
3. Suppose that W_t is a standard Brownian motion, and let $c > 0$. Define $X_t = W_{c+t} - W_c$. Show that $\{X_t, t \geq 0\}$ is a standard Brownian motion that is independent of $\{W_t, 0 \leq t \leq c\}$.
4. Define $Y_t = (1-t)W_{\frac{t}{1-t}}$ for $0 \leq t \leq 1$, compute $\text{Cov}(Y_s, Y_t)$ for $0 \leq s, t \leq 1$. (Y_t in fact is a Brownian bridge)
5. Use the Ito's formula to find the SDE satisfied by each of the following stochastic processes:
 - (a) $X_t = \arctan(t^2 + W_t)$.
 - (b) $X_t = \cos(e^{W_t})$.
6. Calculate the mean and variance of the Ornstein-Uhlenbeck process

$$X_t = \theta(1 - e^{-\mu t}) + X_0 e^{-\mu t} + \sigma \int_0^t e^{\mu(s-t)} dW_s$$

by using the *martingale property* and the *Ito Isometry* of Ito process.

7. Assume that $f(t)$ is a function of time t only and is differentiable, prove that

$$\int_0^T f(t) dW_t = f(T)W_T - \int_0^T W_t f'(t) dt.$$

8. Find the solution of the following SDEs:

- (a) $dX_t = \mu X_t(\theta - \log X_t)dt + \sigma X_t dW_t$,
- (b) $dX_t = \mu X_t dt + (\gamma + \sigma X_t) dW_t$,
- (c) $dX_t = \frac{k}{2} X_t^{2k-1} dt + X_t^k dW_t$, k arbitrary,
- (d) $dX_t = \frac{1}{2} e^{2X_t} dt + e^{X_t} dW_t$,